Transformer-Based Fake News Detection Using DistilBERT

Final Project Report — Experimental implementation, evaluation, deployment and limitations

1. Executive Summary

This project develops a binary text-classification system for fake-news detection using a fine-tuned DistilBERT Transformer model. The system classifies an input news article as REAL or FAKE based on patterns learned from a labeled news dataset. A traditional TF-IDF + Logistic Regression model was also trained on the same train/validation split to provide a baseline for comparison.

The working dataset initially contained 44,898 rows: 23,481 fake articles and 21,417 real articles. After removing exact duplicate article texts, 38,646 unique articles remained. The final modeling input was title + article body; subject and date were deliberately excluded because subject categories showed strong label separation and could create a shortcut rather than genuine content-based detection.

2. Problem Statement

Online news can contain misleading or fabricated information. Manually checking large volumes of text is difficult and time-consuming. The project therefore investigates whether a Transformer-based language model can learn textual patterns associated with fake and real news and classify unseen articles automatically.

3. Objectives

- Prepare and clean a labeled fake/real news dataset.
- Remove exact duplicate article texts and investigate possible data leakage.
- Build a content-based classifier using title + article body.
- Fine-tune DistilBERT for binary classification.
- Build a TF-IDF + Logistic Regression baseline using the same split.
- Compare accuracy, precision, recall and F1-score.
- Create a confusion matrix and inspect prediction behavior.
- Save and reload the trained model from Google Drive.
- Build an interactive Gradio interface for user-entered news.
- Document limitations, especially near-duplicate overlap and the difference between classification and fact-checking.

4. Dataset and Data Preparation

Original files and counts:

- Fake.csv: 23,481 rows.
- True.csv: 21,417 rows.
- Combined: 44,898 rows.
- After exact duplicate removal: 38,646 unique articles.
- REAL label = 0; FAKE label = 1.

The final model dataframe contained only two columns: content and label. content was constructed as title + text. The date and subject fields were excluded.

5. Experimental Split

The primary experiment used an 80/20 stratified split with random_state=42. The resulting primary split contained 30,916 training articles and 7,730 validation articles. The validation set contained 4,239 REAL and 3,491 FAKE articles.

6. Transformer Model

The chosen model was distilbert-base-uncased with DistilBertForSequenceClassification and num_labels=2. The model contains 66,955,010 parameters in this configuration. Text was tokenized with DistilBertTokenizerFast and truncated/padded to max_length=256.

The model was fine-tuned for two epochs with AdamW and learning rate 2e-5. The batch size was 16. The completed training run took approximately 14.2 minutes per epoch on the Colab T4 environment.

7. Primary DistilBERT Results

Metric	Logistic Regression	DistilBERT
Accuracy	98.51%	99.94%
Precision	98.90%	99.86%
Recall	97.79%	100.00%
F1-score	98.34%	99.93%

DistilBERT made 5 validation errors in the primary validation set: 4,234 REAL articles were correctly classified as REAL, 3,491 FAKE articles were correctly classified as FAKE, 5 REAL articles were classified as FAKE, and 0 FAKE articles were classified as REAL.

8. Logistic Regression Baseline

A TF-IDF vectorizer with max_features=50,000, word and bigram features (ngram_range=(1,2)), and English stop-word removal was fitted only on the training texts. Logistic Regression was then trained on the same split. It achieved 98.51% accuracy and 98.34% F1-score.

9. Comparison and Interpretation

On this validation split, DistilBERT achieved higher accuracy, precision, recall and F1-score than the TF-IDF + Logistic Regression baseline. The result supports the experimental use of a Transformer for this dataset. However, the comparison is specific to this dataset and split; it does not establish that DistilBERT will always outperform Logistic Regression on every fake-news dataset.

10. Confusion Matrix

DistilBERT confusion matrix (rows = actual, columns = predicted):

```
                Predicted
              REAL      FAKE
Actual REAL    4234         5
Actual FAKE       0      3491
```

The matrix shows zero false negatives for the FAKE class on this validation set. This is a dataset-specific validation result, not a guarantee for future real-world articles.

11. Near-Duplicate Audit

Because 99.94% accuracy is unusually high, a TF-IDF cosine-similarity audit was performed. The maximum similarity between each validation article and a training article was examined. There were 184 validation articles with similarity >=0.90, 153 with similarity >=0.95, and 111 with similarity >=0.99.

Among the 111 >=0.99 pairs, 107 were REAL-to-REAL, 3 were FAKE-to-FAKE, 1 was FAKE-training to REAL-validation, and none were REAL-training to FAKE-validation. This indicates that near-duplicate overlap exists and should be considered when interpreting the random-split result.

A similarity-aware split was constructed using groups so that highly similar articles were kept together. The robust split contained 30,919 training articles and 7,727 validation articles. A fresh DistilBERT training run on this split was planned but could not be executed because the Google Colab GPU quota was exhausted. Therefore, no robust-split performance number is claimed.

12. Prediction System

The saved model was successfully reloaded from Google Drive and used for CPU inference. The prediction pipeline tokenizes the input, sends it through DistilBERT, applies softmax to the two logits, and reports REAL and FAKE probabilities.

Example test: the sentence 'Scientists Discover That Drinking Two Cups of Coffee a Day Can Increase Human Lifespan by 50 Years' was classified as FAKE with 15.62% REAL probability and 84.38% FAKE probability. This is a model prediction, not independent factual verification.

13. Deployment Interface

A Gradio interface was created so a user can paste an article into a browser-based text box and receive a prediction and probabilities. The interface runs inference using the saved DistilBERT model.

14. Limitations

- The system is a text classifier, not an independent fact-checker.
- The model learns patterns from the dataset and may learn dataset-specific source or writing cues.
- The random validation split contains near-duplicate overlap; therefore the 99.94% score should not be interpreted as real-world accuracy.
- A stricter similarity-aware retraining experiment was not completed because of the Colab GPU quota.
- The model truncates each input to 256 tokens, so information later in very long articles may be ignored.
- Softmax confidence is model confidence, not a calibrated probability that the underlying claim is false.
- The dataset represents a particular collection of news articles and may not represent current news or every domain.

15. Future Scope

- Complete and report the similarity-aware robustness experiment.
- Use grouped or source-aware splitting to reduce dataset-specific leakage.
- Evaluate on a completely external, time-separated or source-separated dataset.
- Calibrate confidence scores.
- Add explainability methods such as token attribution or attention analysis, with appropriate caveats.
- Add source retrieval and external fact-checking as a separate component if the project is extended into a fact-checking system.
- Deploy the Gradio prototype as a standalone web application.

16. Conclusion

The project successfully demonstrates a complete Transformer-based fake-news classification pipeline: data preparation, duplicate removal, stratified splitting, DistilBERT fine-tuning, evaluation, comparison with Logistic Regression, model persistence, inference and a browser-based interface. On the primary random validation split, DistilBERT achieved 99.94% accuracy and 99.93% F1-score, outperforming the TF-IDF + Logistic Regression baseline. The project also explicitly identified near-duplicate overlap as a limitation, which should be considered when interpreting the unusually high validation performance.

17. Suggested Project Architecture

Input News Article → Text Construction/Normalization → DistilBERT Tokenizer → DistilBERT Encoder → Sequence Classification Head → Softmax → REAL/FAKE + confidence → Gradio UI.